

INSURANCE FRAUD ANALYSIS

Insurance fraud remains a persistent and costly issue in Australia's general insurance sector, with car insurance fraud being among the most prevalent types. According to the Insurance Council of Australia, fraudulent claims cost insurers hundreds of millions of dollars each year, leading to increased premiums for honest policyholders and putting pressure on the industry's resources. As a result, Australian insurance companies are increasingly investing in data-driven approaches to detect and prevent fraud more effectively.

The case study is based on dataset of historical motor insurance claims in the Australian context, containing both fraudulent and non-fraudulent records. This dataset includes a diverse range of variables such as vehicle characteristics, driver demographics, accident details, policy information, and more. By analyzing this data, the goal is to identify patterns and indicators that may signify fraudulent activity. Such analysis would not only improve operational efficiency but also contribute to a fairer insurance system by reducing financial losses and supporting legitimate claimants. This analysis is particularly relevant given the increasing importance of artificial intelligence and machine learning in Australia's financial and insurance sectors.

In this case study, there are two tasks. First, you are required to conduct an exploratory data analysis (EDA) using Power BI on a dataset of historical car insurance claims in Australia. Your goal is to identify and compare patterns associated with genuine vs. fraudulent claims and generate actionable insights for the insurance industry to improve fraud detection and policy management. You should follow the principles of good data visualization discussed in the lecture. In the second task, you are required to conduct predictive analysis using RapidMiner to predict the fraudulent claims. Use of any automated functions and tools to create visuals or predictive models is not permitted.

1. Based on these tasks, please submit the following three files: Report File (Complete the report template below, submission file format: “.docx or .pdf”)

2. Power BI Visualization File (.pbix)

3. RapidMiner File (.rmp)

To name the file, your studentid, “s1234567_report” , “s1234567_powerbi.pbix”, “s1234567_rapidminer.rmp) Use of any generative AI tools is strictly prohibited and will be handled as per the course and UQ guidelines for violation of academic integrity and student conduct.

Report Template

Report Requirements

1. Executive Summary: An executive summary is a highlight of the key information in the document and is written for a businessperson. It should contain a clear, well-structured, evidence-based summary of your findings. Provide a clear, concise summary of the key findings and insights derived from both dashboards (part I) and predictive analytics (part II). Summarize how these insights can assist insurance industry to on

what to consider in order to effectively address the problem of Insurance fraud detection (max. 250 words)

This report presents a dual approach to analyzing and detecting car insurance fraud in Australia using Power BI for exploratory data analysis and RapidMiner for predictive modeling. Power BI visualizations reveal that genuine claims are commonly associated with younger vehicles, male drivers, and moderate repair costs, while fraudulent claims often involve expensive vehicles, specific collision types (e.g., side-swipes), and high repair costs. These insights highlight behavioral and financial patterns insurers can use to enhance detection protocols. RapidMiner analysis compared Decision Tree and Random Forest models using cross-validation. The Decision Tree achieved 54.02% accuracy but identified only one fraudulent claim, with a fraud recall of 0.02%. Random Forest achieved 52.00% accuracy, identified 940 fraudulent claims, and delivered a fraud recall of 19.47% and precision of 44.93%. Although Random Forest detected more fraudulent claims, its low recall indicates that further model improvement is needed before practical implementation. This report recommends implementing real-time flagging systems, secondary review triggers, and dynamic pricing strategies based on risk indicators. Ethical considerations such as data bias, transparency, and privacy are also addressed to ensure responsible AI deployment. Overall, this report demonstrates how integrating analytics tools can empower insurers to proactively identify fraudulent activities, enhance claim integrity, and improve operational efficiency.

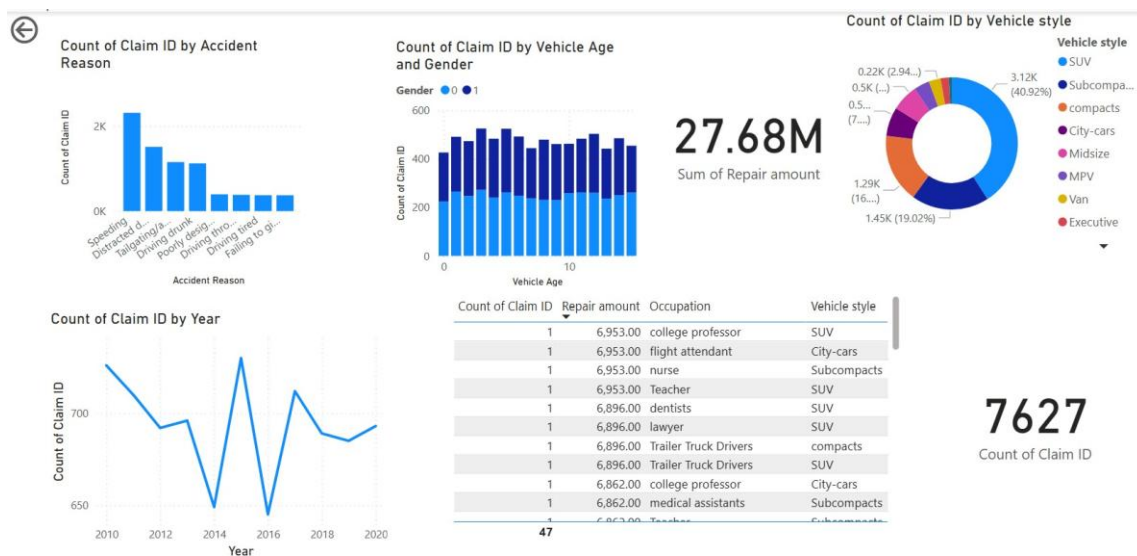
Part I: Power BI Assignment

Please make sure to carefully read and answer the questions. Answering different questions or not answering each part of the questions will lead to a deduction of points. For each section:

2.1 Data Transformation: As we have discussed in the lectures, data are not always in the best shape or quality, and suitable data transformations can reveal interesting relationships. Further, data transformation is also vital for identifying data quality issues. Please report any data transformation (data pre-processing) you conducted before visualising data (max. 150 words).

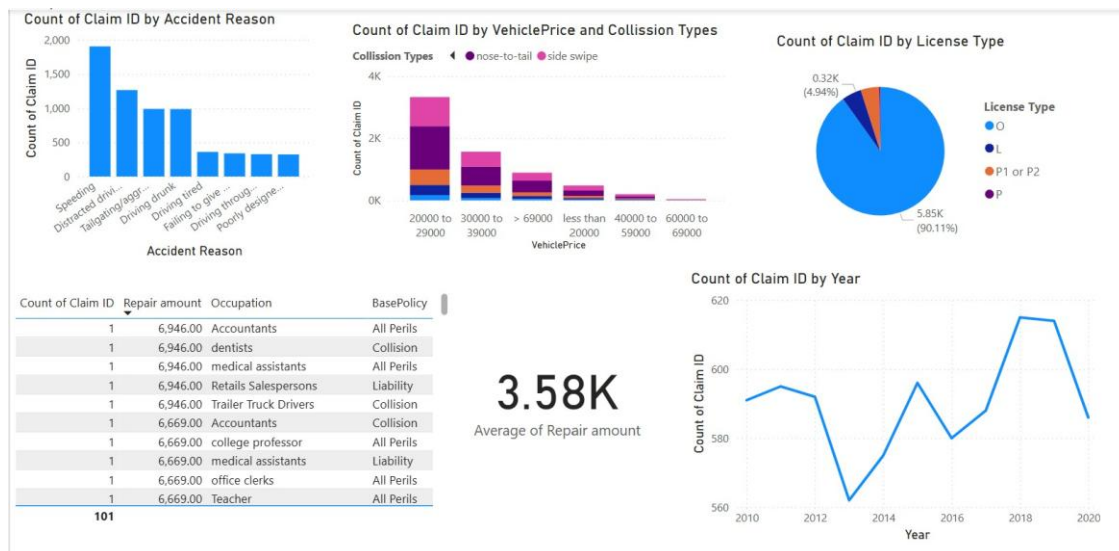
Before importing the dataset into Power BI, I did some transformation steps to improve data quality. For the text-based numeric fields, such as Repair amount and Sum Insured, were cleaned by removing the dollar signs(\$) and converted to numeric types. I also created a new column (Days_Between_Claim_And_Accident) to calculate the number of days between the accident and the claim date. Next, I also converted Date fields into proper datetime format (xx/xx/xxxx), and reviewed the missing values in Marital Status (filtered the NA). Last, Categorical variables such as VehiclePrice, Days_Policy_Claim, and PastNumberOfClaims were standardized to enhance comparison and visualization. By doing these transformation steps and preparing the Cleaned_Insurance_Claim_Date Excel dataset, I can ensure it is clean, structured data and ready for further dashboard analysis.

2.2 Genuine Claims Analysis: Please, prepare a Power BI dashboard that presents the visual analysis of the insurance claims, and the factors related to fraudulent vs. genuine claims to highlight insights from the data. Please use different types of visuals to compare, show relationships, composition, and distributions (See Appendix) between the variables of your choice. Your dashboard should convey main insights, should have a story, and use different types of visuals. The dashboard report should include between 5–8 items (at least 4 visuals, and may include tables and tiles/cards, and other elements).



Here are some visuals that highlight key insights into genuine claims. The bar chart shows that "Speeding" and "Distracted driving" are the top accident reasons. The clustered bar chart shows that vehicles aged between 1–5 years have higher claim volumes, and male drivers have more claims across all vehicle ages. The donut chart shows that the vehicle style - SUV has the highest claim submissions, which is about 40.92%. The line chart shows that claim frequency is highest in 2015, and a card displays the total repair cost of \$ 27.68 M. Also, the table reveals that Trailer Truck Drivers and college professors appear frequently as claimants. This combination of visuals in the dashboards supports strong comparison, composition, and distribution analysis for genuine claim analysis.

2.3 Incident Analytics: Please, prepare a Power BI dashboard that presents the visual analysis of fraudulent insurance that highlights insights from data. Please use different types of visuals to compare, show relationships, composition, and distributions (See Appendix) between the variable of your choice. Your dashboard should convey main insights, should have a story, and use different types of visuals. The dashboard report should include between 5-8 items (at least 4 visuals, and may include tables, and tiles).



The dashboard shows key fraud-related patterns with visuals. The first bar chart shows that “Speeding” and “Distracted driving” are the leading accident reasons of fraudulent claims. The stacked column chart indicates that mid-price range vehicles (around \$20,000–\$29,000) are relevant with more fraud, especially involving “nose-to-tail”(purple) and “side-swipe” (pink) collision types. Also, the donut chart highlights that most fraud cases involve holders of full licenses (“O” light blue). The last line chart reveals a spike in fraudulent claims around 2018 (around 615). Last, the card shows the average fraud repair amount is \$3.58K, while the table shows that the repair costs rate is relatively high (\$6669) and shows occupations like medical assistants, accountants, and teachers appear multiple times. Together, these visuals offer a multi-dimensional view of fraudulent incidents and allow insurers to spot red flags, design better rules, and improve predictive models.

2.4 Discussion of the Results and Conclusions: After conducting a thorough visual analysis of the Insurance claim data, please summarize your main findings. Using the insights gleaned from data visualization and the PowerBI report, please suggest specific key factors that help executives understand the dynamics of the vehicle insurance claims and make three suggestions to help improve/expand business operations. Additionally, you may refer to relevant academic articles, blogs, news sources, and other publications to support your suggestions in an academic writing style, complete with appropriate referencing.<< Develop explanations and compelling recommendations based on the visuals and the reported findings using academic styles of clear, structured, and concise writing not exceeding 200 words >>

The visual analysis has key factors from genuine claims and fraudulent claims. Genuine claims are common among SUV drivers, with speeding and distracted driving as top causes. Most involve full license holders, and total repair costs reach \$ 27.68 M. The fraudulent claims often involve mid-priced vehicles (\$20k–\$29k), nose-to-tail or side-swipe collisions, occupations like accountants or medical assistants and average fraud repair cost is \$3.58K.

If insurers want to improve business operations, they should:

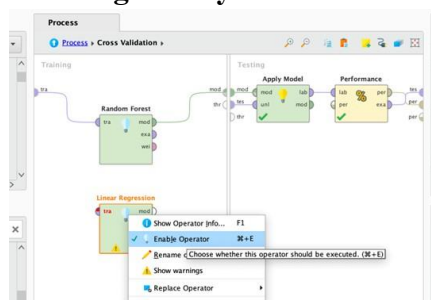
- Ngai et al. (2011) state that they can implement real-time flagging systems to combine high-risk vehicle price ranges, collision types, and driver profiles. This can help obtain useful, non-explicit information from data stored in large repositories.
- According to University Libraries, Virginia Tech (n.d.). They can introduce secondary review protocols for repair costs exceeding defined thresholds. This can help reduce the need for ad hoc decision-making mid-review, and make it an invitation for bias!
- The article from Kuhn & Johnson mentions that leverage predictive modeling and risk segmentation to dynamically price policies and allocate investigative resources efficiently.

These visual and data-driven insights support enhanced fraud detection, operational efficiency, and policyholder fairness through targeted analytics and risk-based strategies.

Part II: RapidMiner Assignment

In this part, students must use RapidMiner to conduct a predictive analysis of insurance claim fraud based on a dataset from an Australian insurance company. You are required to build and compare two supervised machine learning models (e.g., Decision Tree, Logistic Regression, and Random Forest) to whether the claim is fraudulent or not. You can also use any other models as long as you are familiar and know how to use and interpret correctly but report the two best performing models. The aim is to help insurers identify high-risk claims early and improve fraud detection efficiency. If needed, change the “file format” to “CSV”. Also, use RapidMiner to split data into “train set” and “test set”.

3.1 Modelling: Based on the lectures and tutorials, develop and evaluate two different models for classifying claims (fraudulent vs. genuine). You are expected to switch between models in RapidMiner by connecting the respective operators and temporarily disabling those you do not wish to run at the moment (i.e., uncheck “enable operator”):

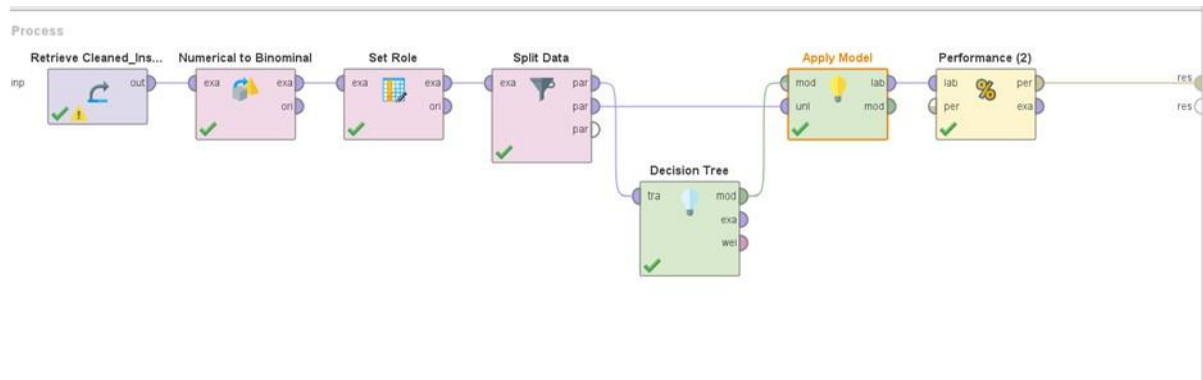


After accessing the process, check whether the process is properly set up and make corrections if necessary. Please document all changes and specifications below. Note: You can choose the models learned from this course or choose any other model outside the tutorial exercise.

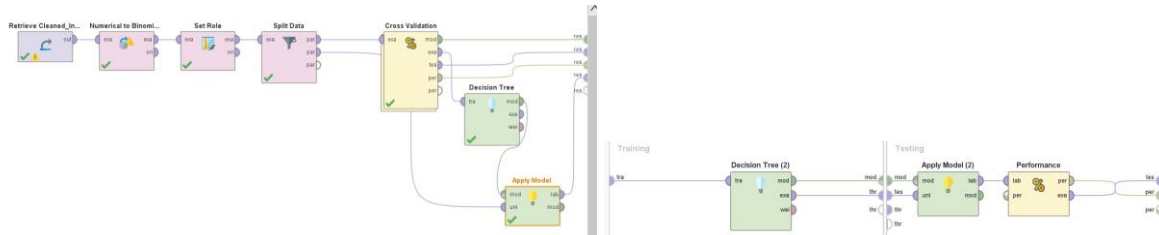
i. Deliverable: Screenshot of Analytical Process

<< Insert screenshots of the analytical process for the first model you choose (include the sub-process within cross validation: one for the process within Cross-Validation and one for the overall Rapidminer process) >>

Decision Tree:

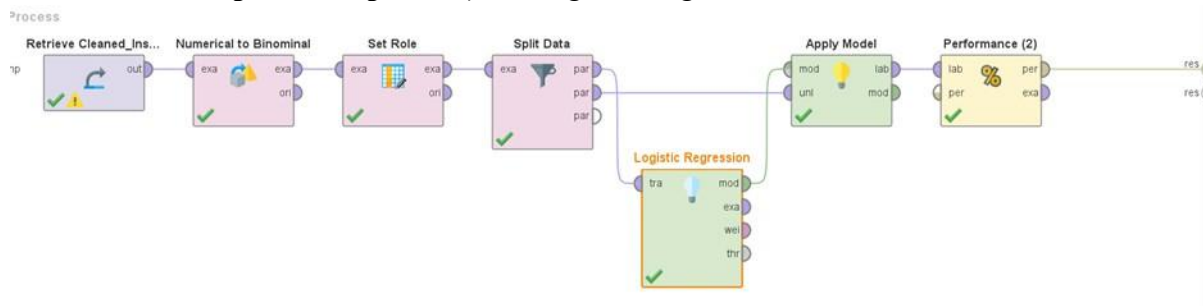


Decision Tree After adding the Cross Validation(Chosen):

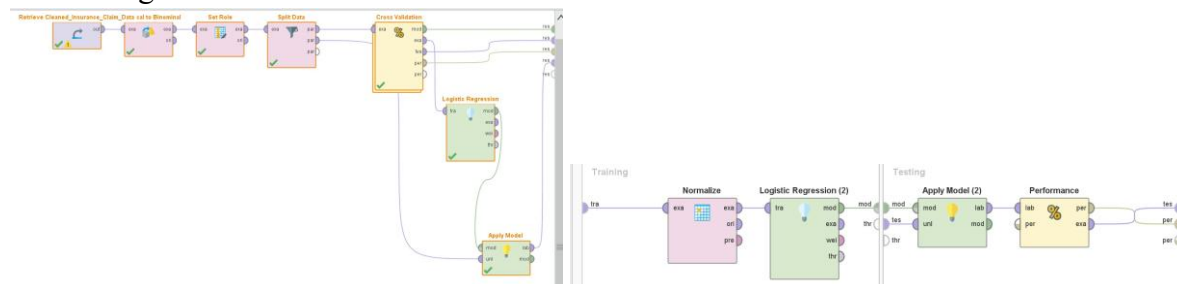


ii. Deliverable: Screenshot of Analytical Process:

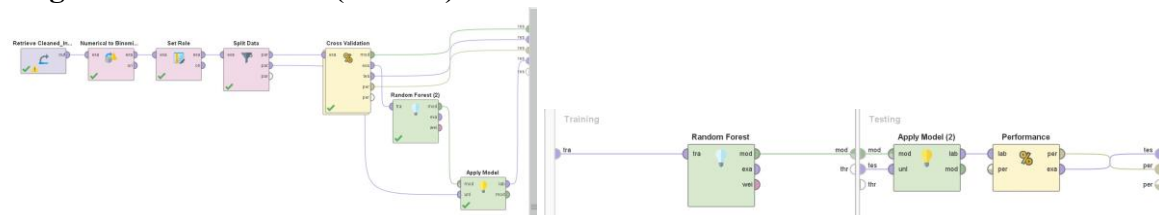
<< Insert screenshots of the analytical process for the first model you choose (include the sub-process within cross validation: one for the process within Cross-Validation and one for the overall Rapidminer process) >> Logistic Regression:



After adding the Cross Validation:



Change to Random Forest(Chosen):



iii.Deliverable: Screenshot of Results of the first model <<

Insert screenshot of the outcomes of the first model>>

Decision Tree:

accuracy: 54.02%

	true false	true true	class precision
pred. false	2431	2069	54.02%
pred. true	0	0	0.00%
class recall	100.00%	0.00%	

Decision Tree After adding the Cross Validation(Chosen):

accuracy: 54.02% +/- 0.04% (micro average: 54.02%)

	true false	true true	class precision
pred. false	5671	4827	54.02%
pred. true	1	1	50.00%
class recall	99.98%	0.02%	

IV.Deliverable: Screenshot of Results of the second model <<

Insert screenshot of the outcomes of the second model>>

Logistic Regression:

accuracy: 46.11%

	true false	true true	class precision
pred. false	19	13	59.38%
pred. true	2412	2056	46.02%
class recall	0.78%	99.37%	

After adding the Cross Validation+Normalize:

accuracy: 46.11% +/- 0.31% (micro average: 46.11%)

	true false	true true	class precision
pred. false	84	70	54.55%
pred. true	5588	4758	45.99%
class recall	1.48%	98.55%	

Logistic Regression change to Random Forest and adding Cross Validation(Chosen):

accuracy: 52.00% +/- 1.01% (micro average: 52.00%)

	true false	true true	class precision
pred. false	4520	3888	53.76%
pred. true	1152	940	44.93%
class recall	79.69%	19.47%	

V . Deliverable: Interpretation of the results of models

Explain the choice and impact of the most decisive predictors in the above models (max. 100 words in total)

In the three models I tried (decision tree, Logistic Regressions, Random Forest), the most decisive predictors were VehiclePrice, Collision Types, and License Type. These key factors helped differ between genuine and fraudulent claims. From all the models, Random Forest achieved the fraud recall (19.47%) with reasonable accuracy. Showing that it is strong at detecting fraud (high recall) with high accuracy, which leads to strong precision and results in an balanced overall performance in future fraud detection and risk modeling strategies.

3.2 Model Comparison: After expertly drawing these conclusions, you anticipate questions regarding the quality of your model. Compare the model performance of both models and justify which method should the management use.

3.2.1 Deliverable: Screenshots of Results of the two models that you chose.

Decision Tree+Cross Validation:

accuracy: 54.02% +/- 0.04% (micro average: 54.02%)

	true false	true true	class precision
pred. false	5671	4827	54.02%
pred. true	1	1	50.00%
class recall	99.98%	0.02%	

Random Forest+Cross Validation:

accuracy: 52.00% +/- 1.01% (micro average: 52.00%)

	true false	true true	class precision
pred. false	4520	3888	53.76%
pred. true	1152	940	44.93%
class recall	79.69%	19.47%	

3.2.2 Deliverable: Interpretation and justification of model performances

Describe the model's performance and which model is better for this analytical task. Justify your preference. You may also refer to the prediction and classification, model selections, and other related academic articles, blogs, and other publications to justify the model selections by following the academic style of writing and referencing (max. 150 words in total).

The Decision Tree after adding cross-validation achieved 54.02% accuracy but failed to detect fraud effectively, which only identified 1 fraud case (recall = 0.02%). In contrast, the Random Forest model with cross-validation achieved slightly lower accuracy (52.00%) but obviously have better fraud detection performance, identifying 940 fraud cases with a recall of 19.47% and precision of 44.93%. We know that fraud detection prioritizes the recall of minority class instances, so Random Forest will be the preferred model. The tree-based ensemble methods like Random Forest are well-suited for imbalancing the datasets and show non-linear feature interactions (Kuhn & Johnson, 2013). While Decision Tree model are simpler and its limited depth are not suitable for complex fraud scenarios. Therefore, the Random Forest provides a more robust and practical solution for improving fraud detection.

3.3 Generate Business Insights and Strategize: Following your analysis of insurance fraud for an Australian car insurance provider, please present a brief reflection on the value of business analytics for the company. You may also refer to the problem of

insurance fraud detection, ethical issues in prediction models, and other related academic articles, blogs, news, and other publications to make recommendations, explain drivers, and discuss ethical issues by following the academic style of writing and referencing. In this reflection, apply your business knowledge to:

3.3.1 Propose specific actions on how to address the identified factors that contribute to insurance fraud detection and suggestions for additional analyses (or data) that could be used to investigate them. (max. 200 words)

To address factors contributing to insurance fraud, I advise that the company should implement automated fraud flagging systems that leverage predictive analytics models. By using Random Forest, they can detect high-risk patterns, for instance, specific VehiclePrice ranges, License Type, and Collision Types. Then this system can prioritize claims for manual review based on the shown fraud probability scores, improve model sensitivity to minority fraud cases, and enhance detection rates without overwhelming false positives.

If they want to have further analysis, they can include socio-demographic variables (income, education level), historical claims behavior, and geographic data to identify more details like regional or behavioral fraud trends. Also, integrating external data sources, such as repair shop activity and police reports, may also help validate and prove the claim authenticity. Moreover, they can apply unsupervised learning techniques (e.g., clustering or anomaly detection). This can help reveal hidden fraud patterns that wasn't captured by supervised models. The routine model performance monitoring and bias audits are also necessary to mitigate ethical concerns such as misclassification or profiling, ensuring fairness and regulatory compliance (Brownlee, 2020; Ngai et al., 2011). By improving in those aspects, these strategies can support a proactive, data-driven approach to fraud prevention.

3.3.2 Now that you have selected the preferred model for this analytical task, discuss the ethical implications that you can identify from the data analytical process? (max. 250 words)

Selecting the Random Forest model as the preferred model for insurance fraud detection helps provide strong technical performance but have some key ethical concerns. A coming major issue will be data bias, if the training data includes systemic inequality or underrepresented groups, the model may unfairly highlight some certain occupations and vehicle types, reinforcing the existing prejudices (O'Neil, 2016; Ngai et al., 2011). Another aspect need to be concern is the risk of false positives, like legitimate claims might be flagged as fraudulent. This will delay claim processing, cause financial stress, and damage customer trust. To solve this, insurers should adopt explainable AI (XAI) frameworks to ensure transparency and accountability in automated decision-making (Ribeiro et al., 2016).

Moreover, continuous monitoring and retraining of models can also help avoiding model drift, maintain fairness, and detect bias early. Integrating a human-in-the-loop process for reviewing high-risk flagged cases further strengthens ethical oversight (European Commission, 2021). Data privacy is also important, the use of sensitive personal information in predictive models must align with legal frameworks such as the Australian Privacy Principles (APPs) and the General Data Protection Regulation (GDPR), ensuring

transparency, consent, and secure data practices (Wachter et al., 2017). In conclusion, balancing predictive power with fairness, transparency, and privacy is essential to maintaining public trust and ethical integrity in AI-driven insurance analytics.

References

Please include all the references (**minimum 10 references (5 per each task)**) that you have cited in your report.

Part 1 (2.4):

Ngai, E. W. T., Hu, Y., Wong, Y. H., Chen, Y., & Sun, X. (2011). The application of data mining techniques in financial fraud detection: A classification framework and an academic review of literature. *Decision Support Systems*, 50(3), 559–569.

<https://doi.org/10.1016/j.dss.2010.08.006>

University Libraries, Virginia Tech. (n.d.). *Systematic reviews & meta-analyses: Protocols*.

<https://guides.lib.vt.edu/SRMA/protocol>

Brownlee, J. (2020). Precision vs recall: Key differences, examples and when to use each. *Machine Learning Mastery*.

<https://machinelearningmastery.com/precision-recall-and-f-measure-for-imbalanced-classification/>

Kuhn, M., & Johnson, K. (2013). *Applied predictive modeling*. Springer.

<https://link.springer.com/book/10.1007/978-1-4614-6849-3> **For Part 2**

(3.3.1) – Model Evaluation and Selection:

Brownlee, J. (2020). Precision vs Recall: Key Differences, Examples and When to Use Each. *Machine Learning Mastery*.

<https://machinelearningmastery.com/precision-recall-and-f-measure-for-imbalanced-classification/>

Scikit-learn Developers. (2023). Classification metrics. https://scikit-learn.org/stable/modules/model_evaluation.html

Ngai, E. W. T., Hu, Y., Wong, Y. H., Chen, Y., & Sun, X. (2011). The application of data mining techniques in financial fraud detection: A classification framework and an academic review of literature. *Decision Support Systems*, 50(3), 559–569.

<https://doi.org/10.1016/j.dss.2010.08.006>

IBM. (2021). What is logistic regression? <https://www.ibm.com/topics/logistic-regression>

Provost, F., & Fawcett, T. (2013). *Data Science for Business: What You Need to Know About Data Mining and Data-Analytic Thinking*. O'Reilly Media.

For Part 2 (3.3.2) – Business Insights & Ethics:

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32.

<https://doi.org/10.1023/A:1010933404324>

Ngai, E. W. T., Hu, Y., Wong, Y. H., Chen, Y., & Sun, X. (2011). The application of data mining techniques in financial fraud detection: A classification framework and an academic review of literature. Decision Support Systems, 50(3), 559–569.

<https://doi.org/10.1016/j.dss.2010.08.006>

O'Neil, C. (2016). Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy. Crown Publishing Group.

Australian Government Productivity Commission. (2017). Data Availability and Use.

<https://www.pc.gov.au/inquiries/completed/data-access/report>

Office of the Australian Information Commissioner. (1988). Privacy Act 1988.

<https://www.oaic.gov.au/privacy/privacy-act>

McKinsey & Company. (2021). Insurance fraud: How analytics is reshaping detection and prevention. <https://www.mckinsey.com>

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 1135–1144).

<https://doi.org/10.1145/2939672.2939778>

Appendix

A.1 Data Dictionary

Feature Name	Description
Claim ID	Unique identifier for the claim
Vehicle style	Type/style of the vehicle (e.g., SUV, compact)
Vehicle color	Colour of the vehicle
Vehicle Age	Age of the vehicle in years

Make	Make/brand of the vehicle
VehiclePrice	Estimated vehicle price range
Gender	Gender of the policyholder (0: Male, 1: Female)
Marital Status	Marital status of the policyholder
State	State in Australia where the claim was made
Occupation	Occupation of the policyholder
License Type	Type of driving license (O, L, etc.)
Drinking-and-Driving	Whether drinking and driving was involved (0: No, 1: Yes)
Collision Types	Type of collision (e.g., side swipe, nose-to-tail)
Accident Reason	Reason for the accident (e.g., speeding, distracted driving)
Holiday Weeks	Whether the accident occurred during a holiday week
AccidentArea	Area type of the accident (Urban/Rural)
Weather	Weather condition during the accident
Accident Date	Date the accident occurred
Claim Date	Date the claim was filed
Police Notification	Whether police were notified (0: No, 1: Yes)
Passengers #	Number of passengers in the vehicle at the time of accident

Repair amount	Amount claimed for repair
Sum Insured	Amount insured for the vehicle
Days_Policy_Claim	Duration of policy before the claim
PastNumberOfClaims	Number of past claims made
AddressChange_Claim	Whether there was a recent address change
BasePolicy	Type of base insurance policy (e.g., Liability, Collision, All Perils)
DriverRating	Driver rating score (1 to 4)
Fraud Detected	Whether the claim was fraudulent (1: Yes, 0: No)

A.2 some suggestions for the visualization choices:

<https://datavizblog.com/wp-content/uploads/2013/04/andrew-abela-chart-chooser-in-color.jpg>